SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 2– Case Study (2 QUESTIONS)

Course Code:	ITA0506	**Course Title:**	COMPUTER VISION
CO Assessed:	CO3 – Precision (BL4)	**Assessment:**	Assessment Tool 1 – Case Study
Weightage:	40% of CIA	**Total Marks:**	20 (2 Questions × 10Marks)
CO1 Statement:	*Analyze and extract image features using detection and matching techniques for effective image representation and comparison..(BL4)*		
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Section / Batch:	SLOT D 2024	**Date:**	11-08-2026

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Code of Conduct

I __Y SRAVAN KUMAR(192472289)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _______SRAVAN__________________

Instructions

- This test contains 2 questions, each carrying 10 marks. Total: 20 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus	Q Nos	Marks
Feature Detection and Matching	Preprocessing, Edge Detection, Harris Corner Detection, Hough Transform, SIFT	1	10
Feature Detection and Matching	PCA, Image Representation, Feature Matching, Image Processing Applications	2	10
		GRAND TOTAL:	**20 Marks**

Annexure A –Case Study

1.A remote sensing application is used to align satellite images of the same geographical region captured at different times for monitoring environmental changes. The system initially relies on edge detection to identify matching regions between the images; however, due to differences in illumination, image orientation, and background details, the matching process becomes unreliable and produces inaccurate alignment results. The development team decides to replace the edge-based approach with Harris Corner Detection to identify stable and distinctive feature points. Analyze the limitations of edge detection in this application and explain how Harris Corner Detection improves feature localization, matching accuracy, and robustness for image registration under varying imaging conditions.

2.An autonomous surveillance system continuously captures high-resolution color images to detect vehicles and pedestrians in real time. Processing the original color images directly increases computational complexity and slows down feature extraction, making it difficult to achieve the required processing speed. To improve system performance, the engineers consider converting the images into grayscale or binary representations before applying feature detection algorithms. Analyze the impact of different image representations on computational efficiency, edge detection, and feature extraction performance, and justify the most appropriate image representation for achieving accurate and reliable object detection in real-time computer vision applications.

Faculty In-charge
Signature: _______________
Date: ___________________

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

Case Study 1: Harris Corner Detection for Satellite Image Registration

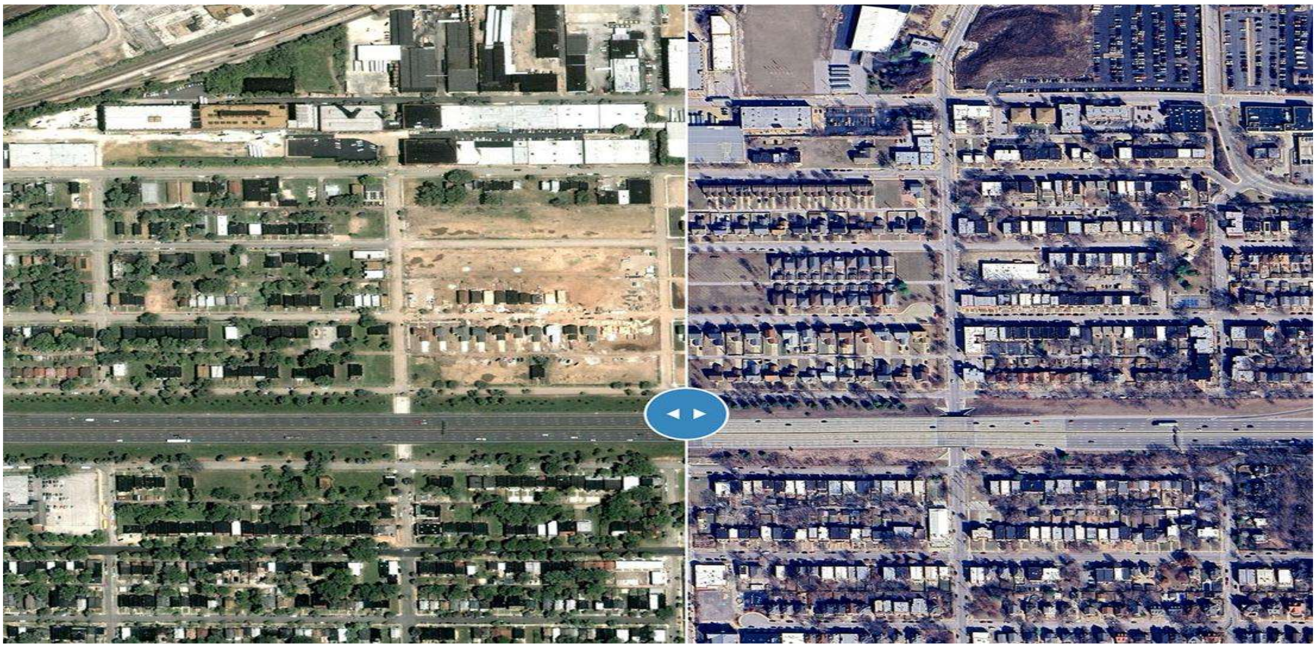
1. Problem

In remote sensing, satellite images of the **same geographical region** are often captured at different times to monitor changes such as:

- Deforestation
- Flooding

- Urban development
- Agricultural changes
- Landslides
- Water-body changes

Before comparing the two images, they must be **properly aligned (registered)**. Even a small registration error can create false changes; studies have shown that sub-pixel misregistration can significantly affect change-detection results.



2. Why Edge Detection is Not Ideal

The original system uses **edge detection** to find matching regions.

Edge detection identifies locations where image intensity changes sharply.

For example:

Satellite Image

↓

Edge Detection

↓

Road boundaries

Building boundaries

River boundaries

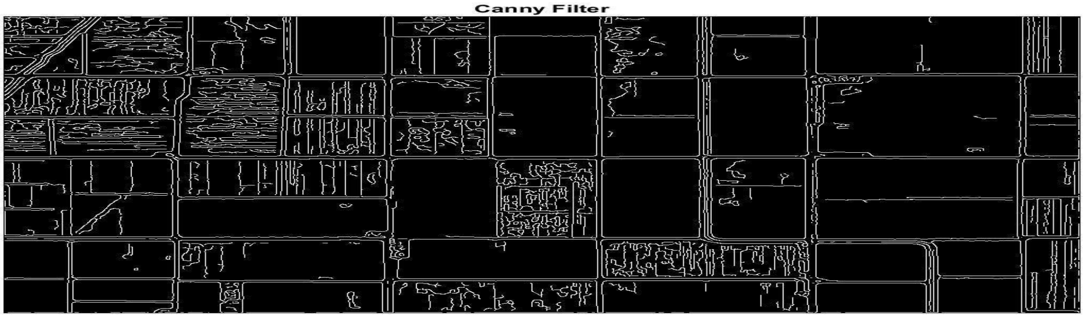
Vegetation boundaries

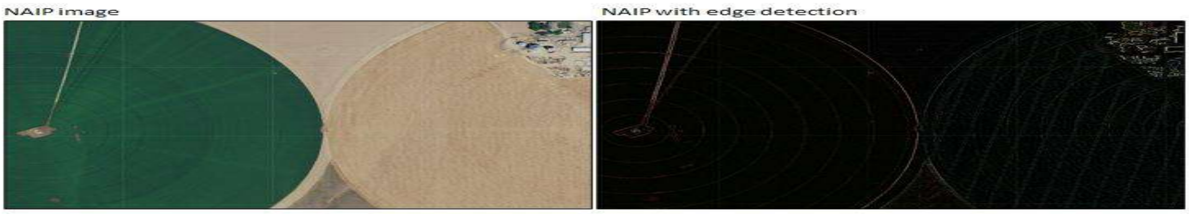
↓

Try to match edges

A satellite image can contain **many edges**, making it difficult to determine which edges correspond to the same physical location.

Canny Filter





Main limitations

1. Illumination changes

Images captured at different times can have different brightness, shadows, and atmospheric conditions.

Therefore:

Same object + different illumination

↓

Different edge response

2. Too many irrelevant edges

Roads, buildings, vegetation, field boundaries, shadows, and background textures can all produce edges.

3. Edges are not distinctive points

An edge is usually a **line or boundary**, rather than a unique location. Matching long edges between two images can therefore be ambiguous.

4. Orientation changes

If the satellite viewing geometry changes, edges may shift or appear differently.

5. Background changes

Seasonal vegetation, clouds, water levels, and construction can alter edges even when the underlying geographic location is the same.

These factors are recognized challenges in remote-sensing change detection, along with image quality, noise, registration errors, illumination differences, and complex landscapes.

3. Harris Corner Detection

The development team replaces edge detection with **Harris Corner Detection**.

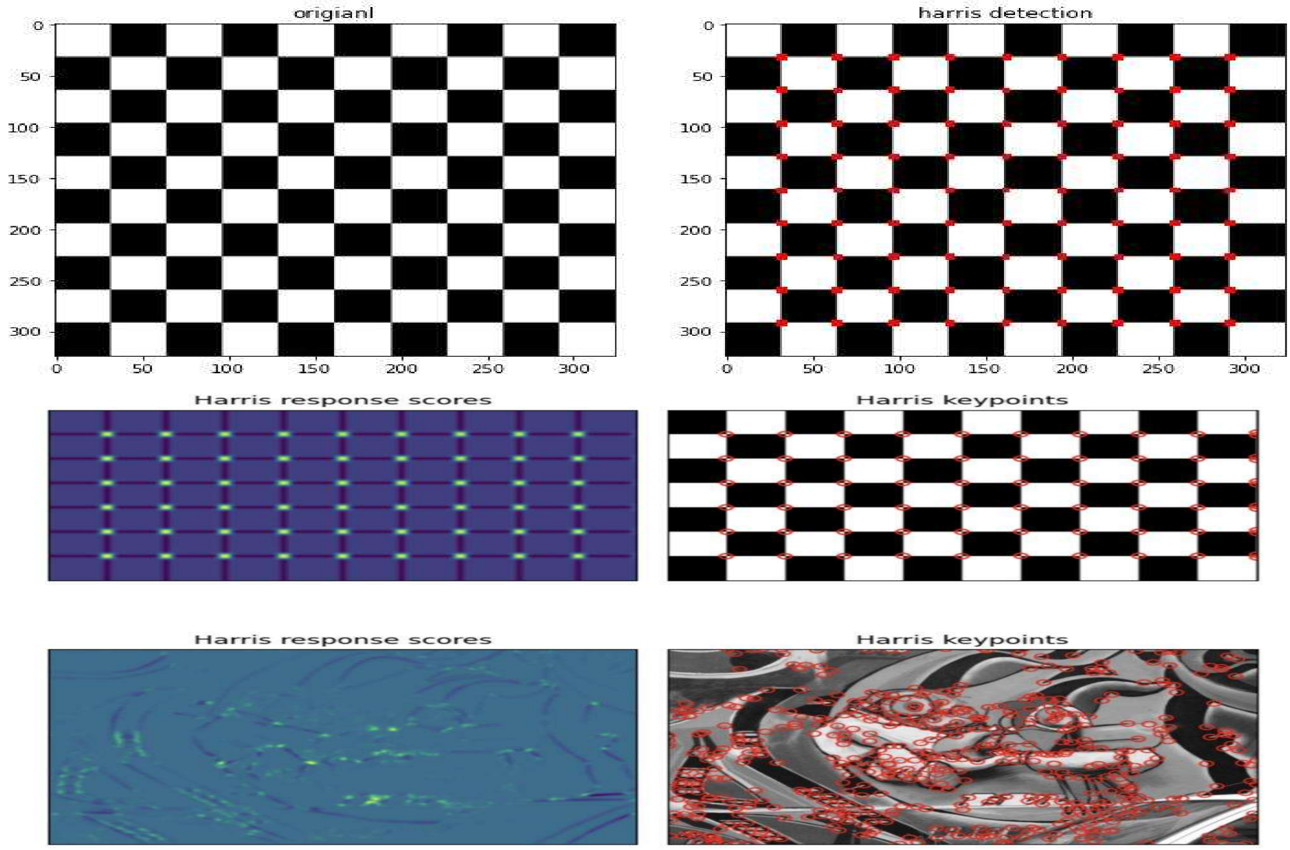
The basic idea is:

A corner is a location where the image intensity changes significantly in **two different directions**.

Harris looks for:

Horizontal change +
Vertical change
↓
Strong change in both directions
↓
Corner

Unlike an edge, which mainly provides information in one direction, a corner gives a more **localized and distinctive feature point**.



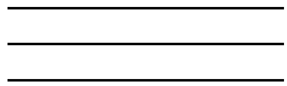
4. How Harris Improves the System

Feature Localization

Harris identifies a **specific point** rather than an extended edge.

For example:

Edge Detection:



Harris:

•

This makes it easier to establish correspondence between the two satellite images.

Better Matching

Suppose the same building appears in two images:

```
Image 1                 Image 2
```

Distinctive corners can act as **reference points** for matching.

Improved Registration

Once corresponding corner points are found:

```
Satellite Image 1
        +
Satellite Image 2
        ↓
Matching Harris Points
        ↓
Estimate Transformation
        ↓
Image Registration
        ↓
Aligned Images
        ↓
Change Detection
```

This is important because accurate registration is a prerequisite for reliable change detection. Modern remote-sensing research continues to treat registration accuracy as a major issue, especially when images come from different dates, sensors, seasons, or viewing conditions.

5. Edge vs Harris

Feature	Edge Detection	Harris Corner Detection
Detects	Boundaries/edges	Corner points
Feature location	Less localized	Highly localized
Number of features	Often very large	More selective
Matching	Can be ambiguous	Easier using point correspondences
Background sensitivity	High	Lower for distinctive corners
Registration	Less suitable alone	Better suited for point-based registration
Computational complexity	Generally low	Moderate
Image matching	Limited	Better

Important qualification

Harris is **not completely invariant** to illumination, rotation, or scale. Its advantage here is mainly that it produces **localized corner points** that are more useful for correspondence than raw edges. For strong scale/rotation changes or substantial illumination differences, modern systems often use descriptors such as **SIFT/SURF/ORB** or learned features rather than Harris corners alone.

6. Overall Solution

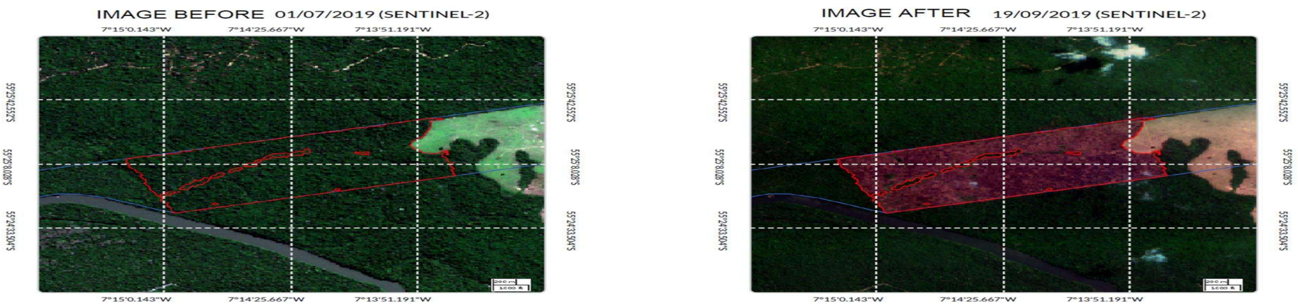
The improved system can be represented as:

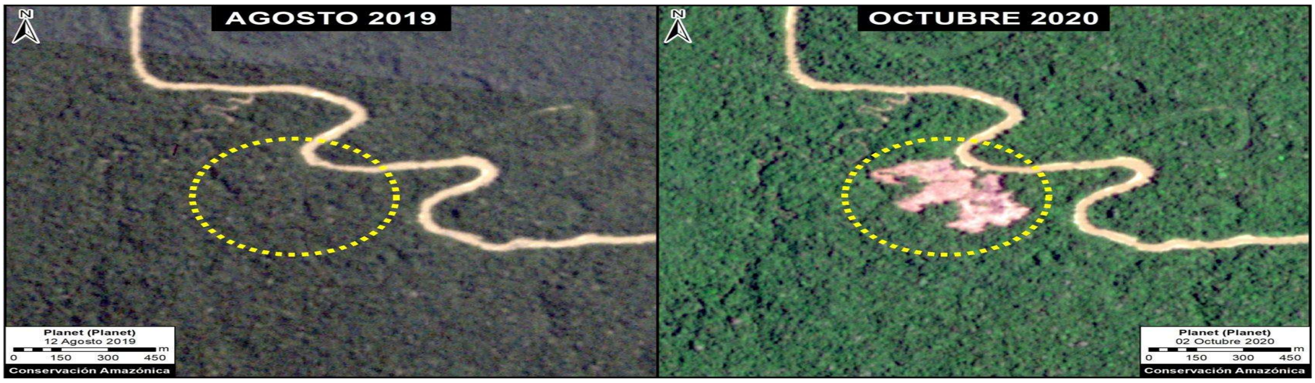
```
Satellite Image - Time 1
        |
        ▼
  Preprocessing
        |
        ▼
 Harris Corner Detection
        |
        ▼
 Corner Feature Points
        |
```

```
        | Matching
        ▼
Corresponding Points
        ▲
        | Matching
        |
Harris Corner Detection
        ▲
        |
  Preprocessing
        ▲
        |
Satellite Image - Time 2
        |
        ▼
  Image Registration
        |
        ▼
   Aligned Images
        |
        ▼
  Change Detection
        |
        ▼
Environmental Monitoring
```

7. Example Application

Suppose a forest region is photographed in **2024** and again in **2026**.





If the images are not aligned:

Misalignment

↓

Apparent differences

↓

False change detection

With better corner-based registration:

Harris Corners

↓

Match corresponding locations

↓

Align images

↓

Compare actual changes

↓

Detect deforestation

8. Conclusion

Edge detection is useful for identifying boundaries, but it can produce many ambiguous features and is sensitive to changes in illumination, background, and viewing conditions.

Harris Corner Detection provides more localized and distinctive points by finding strong intensity changes in two directions. These points can provide better correspondences for image registration and therefore reduce alignment errors before change detection.

Case Study 2: Image Representation for Real-Time Object Detection

1. Problem

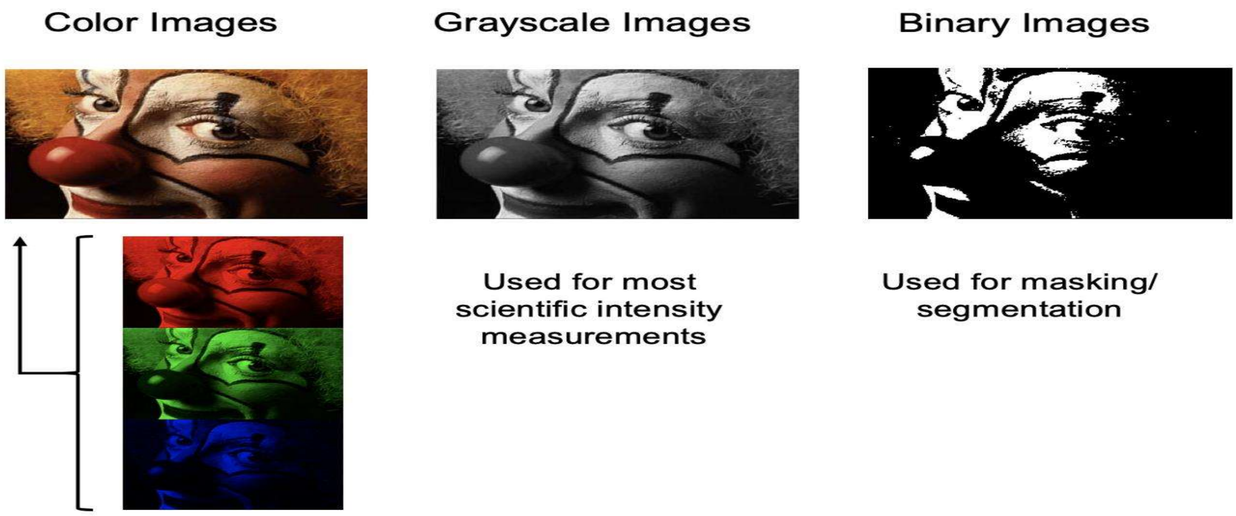
An autonomous surveillance system continuously captures **high-resolution color images** to detect:

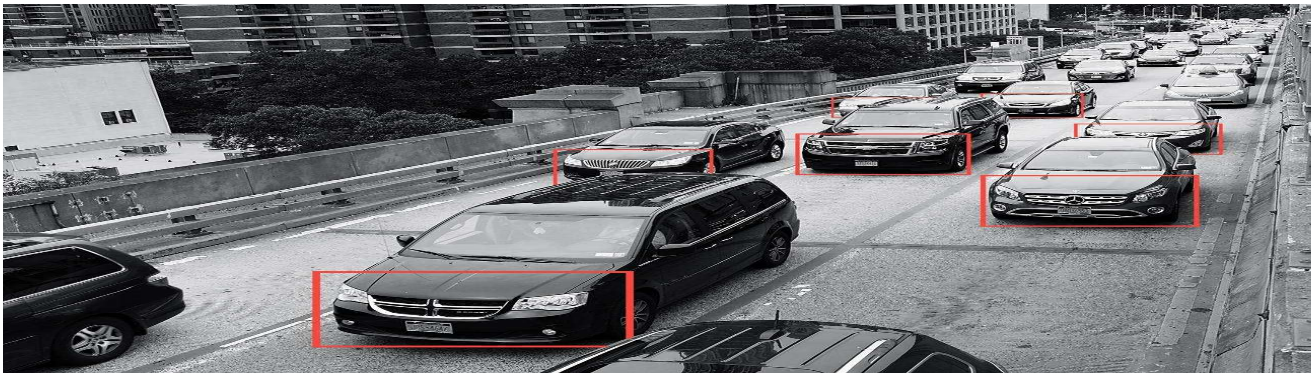
- 🚗 Vehicles
- 🚶 Pedestrians
- 🚦 Traffic-related objects
- 🏢 Other moving objects

Processing full-color images requires more data and computational operations. This can reduce the **frame rate** and make real-time detection difficult.

The engineers are considering three representations:

Color Image
├──► Grayscale Image
└──► Binary Image





2. Color Image

A color image normally contains **three channels**:

R → Red

G → Green

B → Blue

So one pixel contains three intensity values.

Advantages

- Preserves complete color information.
- Useful when color helps distinguish objects.
- Better for applications where object color is important.

Disadvantages

- More data to process.
- Higher memory requirement.
- Feature extraction can be slower.
- More computationally expensive for real-time systems.

For example:

RGB Image
$640 \times 480 \times 3$ channels

The algorithm has to process three channels rather than one.

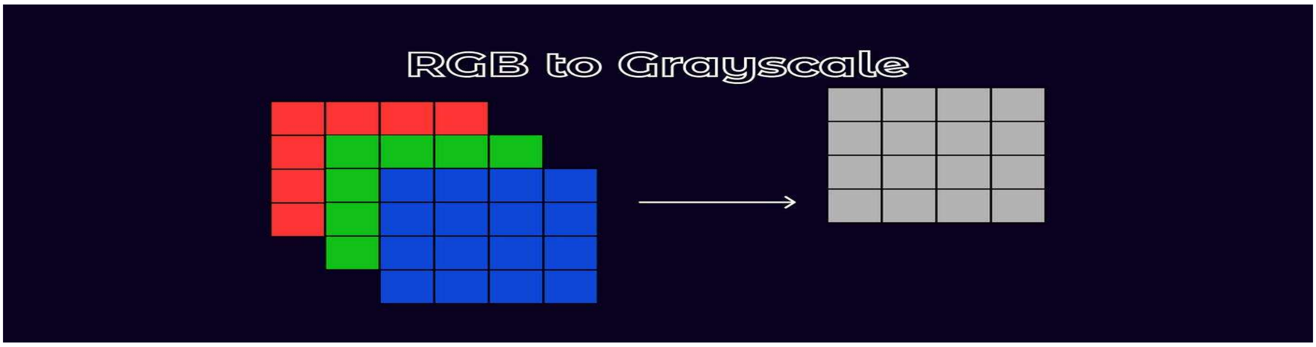
3. Grayscale Image

A grayscale image contains only **one intensity channel**.

Color Image
↓
Grayscale Conversion
↓
One intensity value per pixel

A common conversion is approximately:

$$Gray = 0.299R + 0.587G + 0.114B$$



Original Image

Grayscale Augmentation applied

Advantages

- Requires less data than RGB.
- Faster image processing.
- Lower memory usage.
- Suitable for edge detection.
- Works well with feature detectors such as Harris and SIFT.
- Preserves important structural information such as edges and shapes.

Disadvantage

Color information is lost.

For example, two objects with similar intensity but different colors may become difficult to distinguish.

4. Binary Image

A binary image contains only **two values**:

0 → Black

1 / 255 → White

It is usually produced using **thresholding**.

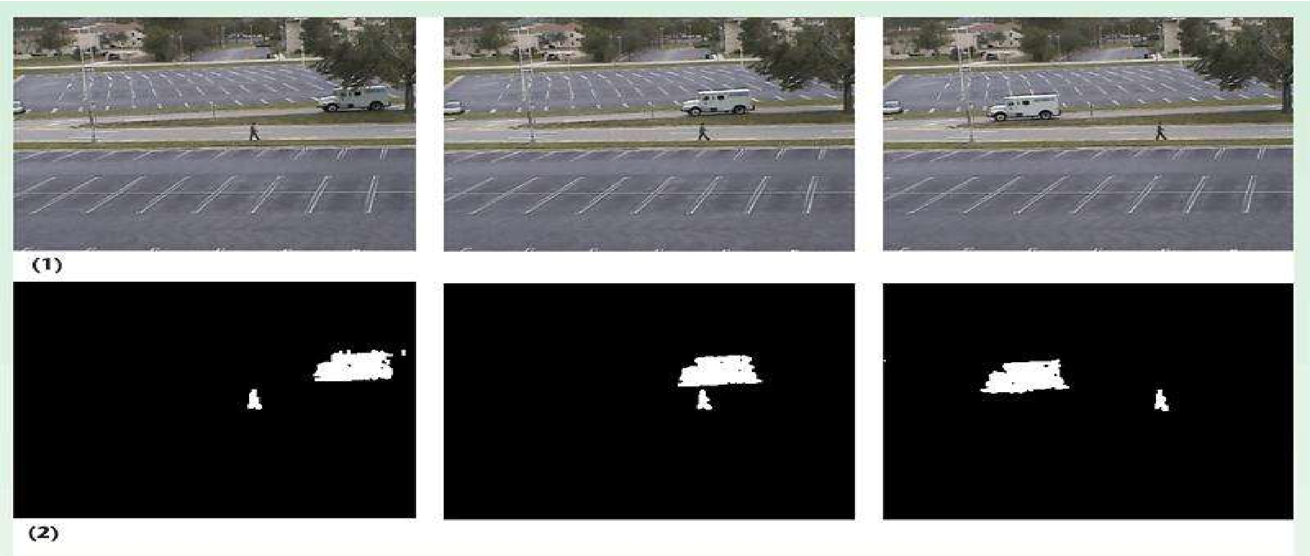
Grayscale Image

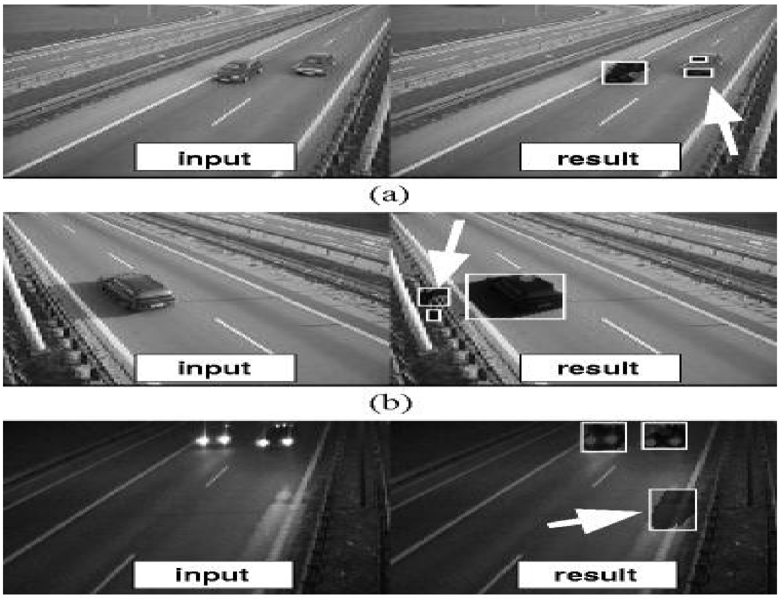
↓

Threshold

↓

Binary Image





(a) (b) (c)

Advantages

- Very low computational cost.
- Very small amount of image information.
- Fast processing.
- Useful for separating objects from a simple background.
- Useful for contour and shape detection.

Disadvantages

- Loses most intensity information.
- Highly dependent on the selected threshold.
- Poor performance under changing illumination.
- Shadows can become unwanted objects.
- Fine object details can disappear.

5. Effect on Edge Detection

Color Image

Edge detection can be performed on multiple channels, but this increases computation.

RGB

↓

R edges

G edges

B edges

↓

Combined result

Grayscale

Usually the best representation for conventional edge detection.

RGB

↓

Grayscale

↓

Canny

↓

Edges

It retains intensity changes while reducing the amount of data.

Binary

Edges can be very clear when the foreground and background are well separated.

However, incorrect thresholding can produce:

Noise

+

Missing edges

+

Broken object boundaries

6. Effect on Feature Extraction

Representation	Feature Extraction	Information	Speed
Color	High computational cost	Very high	Slowest
Grayscale	Efficient	Good structural information	Fast
Binary	Very fast for simple shapes	Low	Fastest

For traditional feature detection:

Grayscale is generally the best compromise.

7. Computational Efficiency

Suppose an image has:

1920 × 1080 pixels

Color

1920 × 1080 × 3

Three channels need to be processed.

Grayscale

1920 × 1080 × 1

Only one channel needs to be processed.

Binary

1920 × 1080

with only two possible intensity values.

Therefore:

Color
↓

More data

↓

More computation

Grayscale

↓

Less data

↓

Faster processing

Binary

↓

Very little information

↓

Very fast processing

8. Which Representation Should Be Selected?

For this **real-time surveillance application**, the best general choice is:

Grayscale

because it provides a good balance between:

- Processing speed
- Memory usage
- Edge detection
- Feature extraction
- Object shape information
- Detection reliability

Binary images are faster, but they can lose too much information and are sensitive to lighting and threshold selection.

Color images contain more information, but processing all three channels increases computational cost.

9. Recommended Pipeline

```
High-Resolution RGB Image
        |
        ▼
  Resize if necessary
        |
        ▼
    Grayscale
        |
        ▼
  Noise Reduction
        |
        ▼
  Edge Detection
        |
        ▼
 Feature Extraction
        |
        ▼
 Object Detection
        |
        ▼
 Vehicle / Pedestrian
     Detection
```

For example:

Camera

↓

RGB Frame

↓

Grayscale

↓

Gaussian Filtering

↓

Canny Edge Detection

↓

Feature Extraction

↓

Object Detection

10. Important Point: Modern Object Detection

There is an important distinction here.

If the system uses **traditional computer vision techniques** such as:

- Canny
- Harris
- SIFT
- HOG
- Contours

then **grayscale is usually an excellent preprocessing representation**.

However, if the system uses modern deep-learning detectors such as **YOLO**, converting everything to grayscale is not automatically better. Modern detectors can use color information as part of their learned representation, so removing color may reduce detection accuracy for some classes.

Therefore, for this particular case study involving **feature detection and real-time processing**, grayscale is the most reasonable choice.

11. Final Comparison

Factor	Color	Grayscale	Binary
Computational cost	High	**Low**	Very Low
Memory usage	High	**Low**	Very Low
Color information	**Yes**	No	No
Edge detection	Good	**Very Good**	Good in simple scenes
Feature extraction	Good	**Very Good**	Limited
Lighting robustness	Better	Good	**Poor**
Shape information	Good	**Good**	Good
Real-time processing	Moderate	**Excellent**	Excellent
Overall suitability	Moderate	**Best balance**	Limited

Conclusion

Grayscale representation is the most appropriate choice for this real-time surveillance system when using conventional image processing and feature-detection techniques. It reduces computational and memory requirements while preserving important structural information such as edges, shapes, and object boundaries. Binary representation provides even faster processing but loses significant image information and is sensitive to illumination and threshold selection. Color representation preserves the most information but increases computational complexity. Therefore, grayscale provides the best balance between computational efficiency and reliable object detection.